Mathematical models

It had long been understood that one of the most basic ecological processes is the struggle for survival: for herbivores to find food, predators to find prey, and prey to avoid being eaten. Predators do everything they can to hunt and eat prey, and the latter do all they can to avoid being eaten. In 1910, Alfred Lotka introduced one of the first mathematical models ever applied to ecology. Now known as the Lotka-Volterra model, its predator–prey equations help predict the population fluctuations of these two groups.

In the early years of the 20th century, Joseph Grinnell conducted extensive research into animals' habitat needs in the western United States. He observed that species had different "niches" within a habitat—and that if two species have approximately the same food requirements, one will "crowd out" the other. Darwin had observed this on his travels aboard HMS *Beagle*, but Grinnell's axiom developed the idea further, as did subsequent research. In 1934, Georgy Gause demonstrated what he called the competitive exclusion principle in laboratory projects. As William E. Odum put it in 1959, "the ecological niche of an organism depends not only on where it lives, but also on what it does."

From field to lab

Laboratory experiments and field observations are the main methods of providing data for the study of ecological processes, but field experiments —in which a local environment is manipulated to test a hypothesis—were not conducted with scientific rigor until Joe Connell's work with barnacles in Scotland. His experiments—the results of which were published in 1961 —were meticulously planned and observed, and were repeatable.

Connell set the "gold standard" for fieldwork, but experiments in laboratories still have a vital role to play, too—as Earl Werner demonstrated 30 years later. His work revealed the nonconsumptive impact of predatory